Paper id: 252776

Roll No: | | | | | | | | | | | | | |

# BTECH
## (SEM IV) THEORY EXAMINATION 2024-25
## COMMUNICATION ENGINEERING

**TIME: 3 HRS** **M.MARKS: 70**

**Note:** Attempt all Sections. In case of any missing data; choose suitably.

## SECTION A

**1. Attempt *all* questions in brief.** **02 x 7 = 14**

| Q no. | Question |
|---|---|
| a. | Represent following time domain signals into frequency domain :<br>(i) δ(t) (ii) Sin(at) |
| b. | Why we need modulation in Communication System? |
| c. | State Carson's Rule for estimating the bandwidth of an FM signal. |
| d. | What is Figure of Merit. |
| e. | Define white noise. |
| f. | Explain pulse amplitude modulation (PAM). |
| g. | Explain QPSK in short. |

## SECTION B

**2. Attempt any *three* of the following:** **07 x 3 = 21**

| | |
|---|---|
| a. | Explain Single Sideband Suppressed Carrier (SSB-SC) modulation. How does it differ from standard Amplitude Modulation (AM)? Derive the mathematical expression for an SSB-SC signal. Discuss its advantages in terms of power and bandwidth. |
| b. | A message signal is given by: $m(t)=5\cos(1000\pi t)$.The carrier signal is given by $c(t)=10\cos(2\pi \cdot 10^5 t)$. Given:For FM: $k_f$=75 kHz/V and For PM: $k_p$=1 rad/V. Write down expression of FM and PM signal. Compare bandwidth of both. |
| c. | Explain following :<br>(i) Random Processes<br>(ii) Autocorrelation Function<br>(iii) Power Spectral Density for White Gaussian noise<br>(iv) Probability Density Function<br>(v) Cumulative Distribution Function |
| d. | Explain the sampling theorem and its significance in the context of pulse modulation. What happens if the sampling rate is below the Nyquist rate? |
| e. | Discuss QAM in detail. |

## SECTION C

**3. Attempt any *one* part of the following:** **07 x 1 = 07**

| | |
|---|---|
| a. | An audio signal given by m(t) amplitude modulates a carrier signal c(t).Where $m(t)=40\sin(600\pi t)$ millivolt and $c(t)=100\sin(2\pi\times10^5 t)$ millivolt. (Load Resistance is of 1ohm). Calculate following:<br>(i) Modulation Index<br>(ii) Equation of AM Signal<br>(iii) Message Signal Power<br>(iv) Carrier Signal Power<br>(v) Total AM Signal Power<br>(vi) Bandwidth of AM Signal<br>(vii) Transmission Efficiency. |
| b. | Explain working of DSB-SC Modulator and Demodulator. |



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**4. Attempt any *one* part of the following: 07 x 1 = 07**

| | |
|---|---|
| a. | Write down difference between NBFM and WBFM in terms of spectral characteristics and mathematical expression. |
| b. | Define angle modulation. Compare frequency modulation (FM) and phase modulation (PM) in terms of signal representation, FOM, and practical implementation. |

**5. Attempt any *one* part of the following: 07 x 1 = 07**

| | |
|---|---|
| a. | Describe how noise affects Amplitude Modulation (AM) systems. Derive an expression for the signal-to-noise ratio (SNR) in a coherent AM receiver. |
| b. | Explain the concepts of pre-emphasis and de-emphasis in FM systems. Derive how these techniques improve the SNR for high-frequency components of the modulating signal. |

**6. Attempt any *one* part of the following: 07 x 1 = 07**

| | |
|---|---|
| a. | Draw and explain block diagram of PCM. Also explain SNR in PCM. |
| b. | Define Delta Modulation and Time Division Multiplexing. |

**7. Attempt any *one* part of the following: 07 x 1 = 07**

| | |
|---|---|
| a. | Explain Phase Shift Keying and Frequency shift keying with help of waveform. Explain working of both. |
| b. | Explain Minimum Shift Keying (MSK) in detail. |

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